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Modeling the cohesive energy and melting point of nanoparticles by their average coordination number

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Abstract

We present a relation between the average coordination number and the cohesive energy for nanoparticles that shows that the ratio of nanoparticles cohesive energy to the bulk value is equal to the ratio of the nanoparticles average coordination number to that of the bulk. We consider the effect of lattice and surface packing factors on the average coordination numbers of the atoms in the nanoparticle. The melting temperature of nanoparticles has been calculated from the obtained relation for cohesive energy, and predictions for the cohesive energy and melting temperature of the nanoparticles have been compared with other theoretical models and available experimental data and the results of molecular dynamics simulations.
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1. Introduction

Nanoclusters have unique properties that result from their very large fraction of surface atoms [1], which strongly influence both their chemical [1–3] and physical properties [4, 5] in comparison with the bulk. Interactions between surface atoms of nanoscale clusters can also lead to significant changes in structure: for example, change in lattice type [6], change in lattice parameters [7], and varying average coordination number for surface and interior atoms [4,8]. The average coordination number is a critical geometrical parameter that depends on the shape and size of a cluster, and many properties of nanoclusters are influenced by it [1,4,8–10]. In many papers, the effect of the coordination number of atoms, especially surface atoms, on the properties of nanoclusters has been addressed [8–14]. Naturally, any decrease in the particle size leads to a decrease in the average coordination number due to an increased number of dangling bonds.

One of the most important properties of a solid that depends directly on the coordination number is the cohesive energy [11], which affects the thermal stability and hence the melting temperature [11–13]. The cohesive energy of a particle is dependent upon its size, and several models discuss this dependency [11–13,15]. The melting temperature of nanoparticles is a basic physical parameter that has a significant impact on properties [16,17], and it is generally recognized that, for free-standing nanoparticles, the melting temperature decreases with decreasing in size [18–20]. There are many models for the size dependency of the nanoparticle melting temperature [13,20–28], and in most of these there is a linear relation between the melting temperature (T_{mp}) and the reciprocal size ($1/D$) [20–25], i.e. $T_{mp}/T_{mb} = 1 - A/D$ where T_{mp} denotes the melting temperature of the bulk material, and A is a constant which is dependent upon the modeling parameters. Generally such a linear relation may not fit experimental observations, especially in the lowest size range.

In the present work, the effects of the packing factors of the lattice and the surface are considered, and a relation is obtained between a nanoparticle's average coordination

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number and its cohesive energy and melting temperature. This leads to a nonlinear equation for the melting point in terms of the reciprocal size of the nanoparticle. The average coordination number calculated in this model is compared with results obtained using the geometrical characteristics of nanoclusters [4]. Our model is compared with the liquid drop model [23,24], and with available experimental data and the results of molecular dynamics.

2. The model

2.1. Average coordination number of a nanoparticle

We define the parameter Z_{Lp} as the coordination number of atoms in the lattice and Z_{Sp} as the surface crystalline planes of the particle. Note that Z_{Sp} is the number of surface bonds per surface atom, without taking into consideration the bonds to interior atoms. Now we define another factor: the average coordination number of the nanoparticle ($\bar{Z}_p$), which is the average of the lattice and surface coordination numbers of atoms in the nanoparticle:

$$\bar{Z}_p = \left(\sum_{k=1}^{n_i} Z_{Lp,k} + \sum_{j=1}^{n_s} Z_{Sp,j}\right) \Big/ n_t \tag{1}$$

where n_s, n_i and n_t, are the number of surface atoms, interior atoms, and the total number of atoms in the nanoparticle, respectively. $Z_{Lp,k}$ and $Z_{Sp,j}$ are the lattice and surface coordination numbers, respectively. We assume that $Z_{Lp,k}$ is equal for all of the interior atoms ($Z_{Lp,k} = Z_{Lp}$) and $Z_{Sp,j}$ is equal for all of the surface atoms ($Z_{Sp,j} = Z_{Sp}$) for a particular nanoparticle size, and $n_t = n_i + n_s$. Therefore:

$$\bar{Z}_p = \frac{n_i Z_{Lp} + n_s Z_{Sp}}{n_i + n_s}. \tag{2}$$

We should note that Z_{Lp} and Z_{Sp} may vary with the size of the nanoparticle. Now we define the fraction of interior and surface atoms, f_i and f_s respectively, as: $f_s = n_s/(n_i + n_s)$, $(f_i + f_s = 1)$, so we can rewrite Eq. (2) as: $\bar{Z}_p = f_i Z_{Lp} + f_s Z_{Sp}$.

We also define a parameter, q as the ratio of the coordination number of atoms on the surface to that in the lattice: $q = Z_{Sp}/Z_{Lp}$, and hence Eq. (2) can be written as:

$$\bar{Z}_p = Z_{Lp}\left(1 - (1-q) f_s\right). \tag{3}$$

To obtain $\bar{Z}_p$ we must calculate the total number of atoms and the number of surface atoms of the nanoparticle, and then calculate the value of f_s for the nanoparticle. The number of surface atoms and total atoms have been calculated in our previous work [27,28], and using them f_s equals to:

$$f_s = \frac{4 P_S r}{P_L R + 2 P_S r}. \tag{4}$$

Here P_S is the packing factor of the surface crystalline planes, defined as the ratio of the surface area of the plane occupied by atoms to the total surface area of the plane; and P_L is the packing factor of the lattice, defined as the ratio of the volume of the unit cell occupied by atoms to the total volume of the unit cell. R is the average radius of the nanoparticle, and r is the atomic radius, deduced from atomic volume ($V_a = 4\pi r^3/3$) [23,24].

We can define a critical size (D_0) for nanoparticles at which all of the atoms are located on the surface, i.e. $n_i = 0$ ($f_i = 0$), or equally $f_s = 1$, ($n_s = n_t$). From Eq. (4) we have: $D_0 = 4r P_s/P_L$. We can rewrite Eq. (4) in terms of D_0:

$$f_s = \frac{2D_0}{D + D_0} \quad (D, \text{the diameter of the nanoparticle} = 2R). \tag{5}$$

Eq. (5) represents f_s as a function of nanoparticle critical size and, by substitution into Eq. (3), we can calculate $\bar{Z}_p$ as a function of nanoparticle size:

$$\bar{Z}_p = Z_{Lp}\left(1 - 2(1-q)\frac{D_0}{D + D_0}\right). \tag{6}$$

We should note that Z_{Lp} is the lattice coordination number of the nanoparticle and can be dependent upon its size. Eq. (6) shows the variation of the particle average coordination number as a function of its size. For large particles close to the bulk ($D \to \infty$), $\mathrm{Lim}_{D\to\infty} \bar{Z}_p$ is the average coordination number of the bulk material. By defining this value as $\bar{Z}_b$ (the bulk average coordination number), i.e. $\bar{Z}_b = \mathrm{Lim}_{D\to\infty} \bar{Z}_p$, from Eq. (6), we also have: $\bar{Z}_b = \mathrm{Lim}_{D\to\infty} Z_{Lp}$ (since the 2nd term in the equation tends to zero). Since $\mathrm{Lim}_{D\to\infty} Z_{Lp}$ is the lattice coordination number of atoms in the bulk material, it can be understood that the bulk average coordination number ($\bar{Z}_b$), is equal to the bulk lattice coordination number (Z_{Lb}), i.e. $\bar{Z}_b = Z_{Lb}$, where $Z_{Lb} = \mathrm{Lim}_{D\to\infty} Z_{Lp}$. For example, for a Face-Centered Cubic lattice (FCC), we have: $\mathrm{Lim}_{D\to\infty} Z_{Lp} = 12$ and therefore $\bar{Z}_b = Z_{Lb} = 12$.

2.2. Cohesive energy and melting temperature of a nanoparticle

To calculate the cohesive energy of a nanoparticle, it is necessary to calculate the total number of the bonds n_B as below:

$$n_B = \frac{1}{2}\left(\sum_{k=1}^{n_i} Z_{Lp,k} + \sum_{j=1}^{n_s} Z_{Sp,j}\right). \tag{7}$$

In Eq. (7), the prefactor 1/2 is required because each interior and surface bond is associated with two atoms. The bonds between the surface atoms and their nearest interior atoms (interior-surface bonds) have not been counted twice in this way, leading to some deviation in the smallest particles. Taking into account this discrepancy leads to a more complex equation which cannot express cohesive energy simply, i.e. in terms of the average coordination number. Using Eqs. (1) and (3), we can rewrite Eq. (7) as:

$$n_B = \frac{\bar{Z}_p n_t}{2} = \frac{Z_{Lp}(n_i + q n_s)}{2}. \tag{8}$$

The cohesive energy of the nanoparticles E_p is the energy required to separate it into discrete atoms. So by considering the

coordination numbers in the lattice and the surface, we have:

$$E_p = \frac{1}{2}\left(\sum_{k=1}^{n_i} \varepsilon_{L,k} \cdot Z_{Lp,k} + \sum_{j=1}^{n_s} \varepsilon_{S,j} \cdot Z_{Sp,j}\right) \tag{9}$$

where $\varepsilon_{L,k}$ and $\varepsilon_{S,j}$ are the energies of each bond in the lattice and the surface, respectively. Here we assume that the bond strength of all atoms of the particle are the same, i.e. $\varepsilon_{L,k} = \varepsilon_{S,j} = \varepsilon$, and rewrite Eq. (9) as:

$$E_p = \frac{1}{2}\varepsilon\left(\sum_{k=1}^{n_i} Z_{Lp,k} + \sum_{j=1}^{n_s} Z_{Sp,j}\right). \tag{10}$$

This assumption is appropriate for pair potentials, such as Lennard-Jones or Morse, although metals cannot be properly described by pair potentials. Also it has been shown that bonds that are related to the low-coordination of atoms of metallic elements are stronger (bond-order/bond-strength correlation) [12,13]. As a result, the assumption of same strength of all atoms of the particle may lead to some deviation in our model. Now by using Eqs. (7)–(9) we have:

$$E_p = \varepsilon n_B = \frac{\varepsilon n_t \bar{Z}_p}{2}. \tag{11}$$

So the cohesive energy of the bulk material (E_b) is: $E_b = \mathrm{Lim}_{D\to\infty} E_p = (\varepsilon n_t/2)\mathrm{Lim}_{D\to\infty}\bar{Z}_p$ and, since $\bar{Z}_b = \mathrm{Lim}_{D\to\infty}\bar{Z}_p$, so we have:

$$E_b = \frac{\varepsilon\, n_t \bar{Z}_b}{2}. \tag{12}$$

Hence, from Eqs. (11) and (12), the ratio of the cohesive energy of the nanoparticle to the bulk can be obtained as:

$$\frac{E_p}{E_b} = \frac{\bar{Z}_p}{\bar{Z}_b}. \tag{13}$$

This shows that the cohesive energy of the nanoparticle varies linearly with its average coordination number. When the size of the nanoparticle increases, $\bar{Z}_p$ tends towards the bulk average coordination number ($\bar{Z}_b$), so $E_p/E_b = 1$.

The melting temperature of a solid is proportional to its cohesive energy [23], and hence by attention to Eq. (13):

$$\frac{T_{mp}}{T_{mb}} = \frac{\bar{Z}_p}{\bar{Z}_b}. \tag{14}$$

By considering Eq. (6), we have:

$$\frac{E_p}{E_b} = \frac{T_{mp}}{T_{mb}} = \frac{Z_{Lp}}{Z_{Lb}}\left(1 - 2(1-q)\frac{D_0}{D+D_0}\right). \tag{15}$$

Eq. (15) shows a general relationship between the size and cohesive energy and melting temperature of the nanoparticles in terms of the critical diameter, D_0 which is dependent upon the lattice and surface packing factors. We should notice that $\bar{Z}_b$ is the bulk average coordination number and is equal to the bulk lattice coordination number (Z_{Lb}), whilst Z_{Lp} is the lattice coordination number of the nanoparticle.

2.3. Average coordination number of nanoclusters from geometrical characteristics

Montejano et al. [4] have presented analytical expressions for the geometrical properties of atomic clusters which are dependent upon the cluster order (ν, the number of its crusts). By this method, they have calculated the average coordination number in the cluster surface and lattice. We assume that relatively large particles ($D > 6$ nm) can be considered as spheres [29], and are therefore approximately equivalent to the nanoclusters described by Montejano et al. [4]. Here we apply the calculated parameters from the work of Montejano et al. [4] to an FCC lattice. The ICO (icosahedral) cluster structure is composed of close-packed planes [4,30], and this is close to our model because we consider that the nanoparticle surface is composed of the closest packed planes ($P_S = 0.91$). In contrast, in the CO (cubo-octahedral) nanocluster structure there exist surface planes with non close-packed arrangements [4,30]. For ICO structures, we have the following equation [4]:

$$\bar{Z} = 6\nu(20\nu^2 + 15\nu + 7)/(10\nu^3 + 15\nu^2 + 11\nu + 3) \tag{16}$$

where $\bar{Z}$ is the average coordination number of the nanocluster. When a crust adds to the nanocluster, the diameter increases by a value of $2d$ (d is the atomic diameter), so we assume that there is a simple relation: $D = d(1 + 2\nu)$. To avoid complexity we have not substituted for D in Eq. (16), but the subsequent plots use this conversion in a variable from ν to D.

3. Results and discussion

In the following, to plot Eq. (15) it is assumed that $Z_{Lp} = Z_{Lb}$, i.e. the particles lattice coordination number is independent of its size. Although this assumption may be valid for sufficiently large particles, this may however lead to some variation from reality for the smallest particles. Additional information for graph plotting is shown in Table 1.

Fig. 1 shows the dependency of the average coordination number ($\bar{Z}_p$) on the particle size for Au, using Eq. (6) (assuming $q = 1/2$), in comparison with Eq. (16), which is from the work of Montejano et al. [4]. This shows that the variation of the average coordination number in our model and Ref. [4] are in a similar form, with a maximum difference of about 0.5. This small difference may be explained by the fact that, in the work of Montejano et al. [4], it has been assumed that the surface of the cluster is composed of planes, edges, and vertices. At small sizes the fraction of edge and vertex atoms is considerable, leading to some variation from our model. Also f_s of the nanoparticle, using Eq. (5) and the work of Ref. [4] are compared, in Fig. 1. Again, this shows good consistency between these two models.

Eq. (15) illustrates that the ratio of the nanoparticle average coordination number to that of the bulk represents an important physical parameter. By replacing $X = 1/D$ in Eq. (15) and using Mclaurin's series about $X = 0$, we have:

$$\frac{E_p}{E_b} = \frac{T_{mp}}{T_{mb}} = 2(1-q)\left(\frac{Z_{Lp}}{Z_{Lb}}\right)\sum_{m=0}^{\infty}(-D_0X)^m. \tag{17}$$

Table 1
Packing factors of surface crystalline planes (P_S) and lattice (P_L), and critical size (D_0) for standard lattice structures: face-centered cubic (FCC), hexagonal closed-packed (HCP), body-centered cubic (BCC), and simple cubic (SC)

Lattice structure	Most compacted crystalline plane	P_S	P_L	P_S/P_L	$D_0 = (4P_S/P_L)r$
FCC	(111)	$\pi/(2\sqrt{3})$	$\pi/(3\sqrt{2})$	$\sqrt{3/2}$	$4\sqrt{3/2}\,r$
HCP	(0001)	$\pi/(2\sqrt{3})$	$\pi/(3\sqrt{2})$	$\sqrt{3/2}$	$4\sqrt{3/2}\,r$
BCC	(110)	$3\pi/(8\sqrt{2})$	$\pi\sqrt{3}/8$	$\sqrt{3/2}$	$4\sqrt{3/2}\,r$
SC	(100)	$\pi/4$	$\pi/6$	$3/2$	$6r$

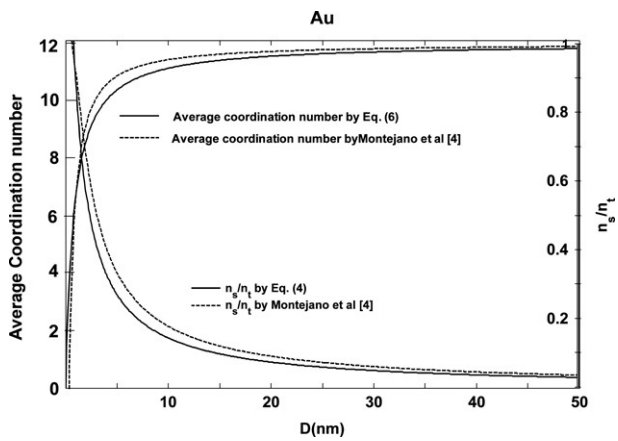

Fig. 1. Dependence of average coordination number ($\bar{Z}_p$) and fraction of surface atoms on nanoparticle diameter for Au. Lattice type of Au is FCC and $D_0 = 0.780$ nm (atomic radius from [23]). Eq. (6), with $q = 1/2$, is compared with the work of Montejano et al. [4].

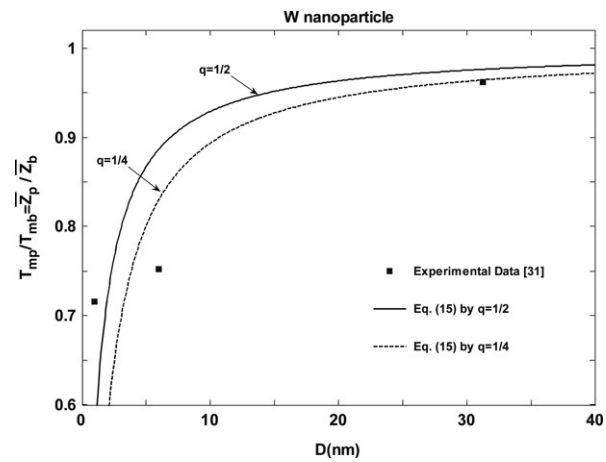

Fig. 2. Dependence of cohesive energy of W nanoparticles on the nanoparticle diameter. Lattice structure of W is BCC, but it can change to FCC by decreasing size. For W, $D_0 = 0.768$ nm (atomic radius from [23]) and $E_b = 8.54$ eV [31]. Experimental data for the cohesive energy of W nanoparticles from [31].

Using this series is appropriate for small values of X $(= 1/D)$, i.e. large particle sizes. If we assume that the surface of the nanoparticle is composed of close-packed crystalline planes (because these planes have a minimum surface energy), then for adequately large particles (close to the bulk), we have $q = 1/2$ (for example, for an FCC bulk material: $Z_{Sb} = 6$ and $Z_{Lb} = 12$). For adequately large particles we can ignore the higher terms of Eq. (17) and, consider only the first and second terms. So by assuming $q = 1/2$ and $Z_{Lp}/Z_{Lb} = 1$, we have:

$$\frac{E_p}{E_b} = \frac{T_{mp}}{T_{mb}} \cong 1 - \frac{D_0}{D}, \quad D_0 = \frac{4P_S r}{P_L}. \tag{18}$$

Eq. (18) is a linear relation in terms of $1/D$ that is true only for relatively large particles, sufficiently close to the bulk. Other models suggest such a linear relation for cohesive energy [11, 15,23,24] and melting temperature [20–25], for the entire range of particle sizes. In our model, A is a linear function of atomic radius (r) ($A = D_0 = 4P_S r/P_L$). Table 1 gives the surface and lattice packing factors, P_S,P_L and the critical size (D_0) for standard lattice structures: face-centered cubic (FCC), hexagonal closed-packed (HCP), body-centered cubic (BCC) and simple cubic (SC). The ratio of P_S/P_L for FCC, HCP, and BCC is the same ($P_S/P_L = \sqrt{3/2}$), and A for all these elements varies in the same way: $A = 4\sqrt{3/2}\,r$. This shows that for FCC, HCP, and BCC structures with the same atomic radius, D_0 is constant. Hence, for an element that can exist with different lattice structures (e.g. tungsten, W), if these structures are HCP or FCC or BCC, and the atomic radius doesn't change, then the critical size D_0 is fixed. For the SC lattice, $P_S/P_L = 3/2$, and therefore D_0 has a larger value in this less compacted structure, leading to a more significant drop in melting temperature with decreasing nanoparticle size.

In Eq. (18), it is assumed that $q = 1/2$, i.e. $Z_S = Z_L/2$, whilst for very small particles we may expect that $q < 1/2$. In our previous work [27], we have shown that the factor of $q = 1/4$ is more suitable for smaller particles, and this value has been used to plot figures for the smaller sizes.

Fig. 2 illustrates the size dependency of the cohesive energy of tungsten (W) nanoparticles. Graphs are plotted from Eq. (15) with two different values of q (1/2, 1/4), and show good consistency with the experimental data from Ref. [31], especially for larger particles. It should be noted that by increasing the size of a nanoparticle, the cohesive energy may become closer to the assumption of $q = 1/2$; however, the experimental data is not sufficient to show this.

The variation of the melting temperature from Eq. (15) is plotted versus the diameter of nanoparticles for Ar, Au, and Sn, respectively in Figs. 3–5, and a comparison is made with the Liquid Drop model [23,24]. From Figs. 3–5 it is found that our model has a good agreement with the experimental data, whilst the Liquid Drop model generally overestimates the size-dependent reduction of the melting temperature.

In Fig. 3, for Ar nanoparticles, our model shows good consistency with molecular dynamics (MD) simulation [32], and illustrates that $q = 1/2$ is appropriate for large particles, whilst $q = 1/4$ gives a better fit at smaller sizes. The lattice structure of Ar in the solid state is FCC, and hence we assumed $P_L = 0.74$ and $P_S = 0.91$ (the surface packing for the most

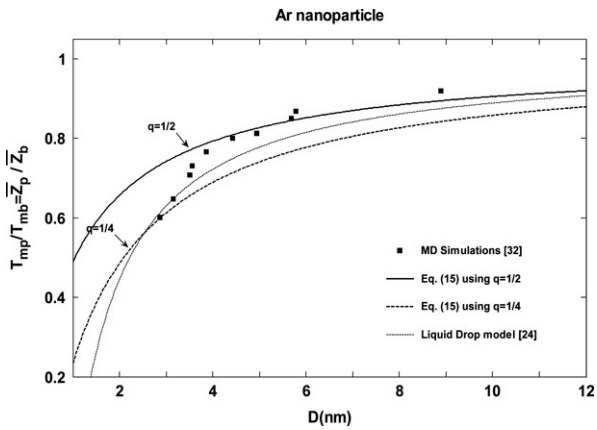

Fig. 3. Dependence of melting temperature on the nanoparticle diameter for Ar. Lattice structure of solid Ar is FCC, $D_0 = 1.041$ (atomic radius from [24]) and $T_{mb} = 84$ K [24]. Molecular Dynamics (MD) simulations for the melting temperature of Ar nanoparticles are from [32].

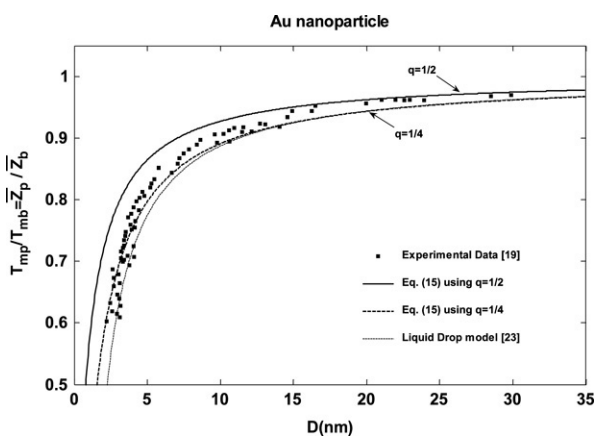

Fig. 4. Dependence of melting temperature on the diameter of Au nanoparticles. Lattice structure of Au is FCC, $D_0 = 0.780$ nm (atomic radius from [23]) and $T_{mb} = 1337.6$ K [23]. Experimental data from [19].

compact crystalline plane (111)). This shows that our model, due to the consideration of the packing factors of lattice and surface, can be regarded as an appropriate one for non-metallic, as well as metallic, materials.

Fig. 4 shows close agreement between the experimental data [19] for Au, the Liquid Drop model [23] and our model by considering the value of $q = 1/4$. Here the Liquid Drop model [23] overestimates the reduction in the melting temperature for larger particles.

Fig. 5 shows the melting temperature of Sn nanoparticles, with a BCT lattice in which the most compacted planes are (100) and (110), with packing factors of $P_{S\,(100)} = 0.71$ and $P_{S\,(110)} = 0.50$. We have plotted the graphs by using an average of $P_S = 0.61$ and $P_L = 0.53$, which gives the best fit with the experimental data [22].

Fig. 5 illustrates that our model, which takes into consideration lattice and surface packing factors, is applicable to other lattice structures besides the simple close-packed structures, and leads to close correspondence with experimental data.

Generally A in our model is lower than that of [23,24], which is the reason why the reduction of melting temperature, for

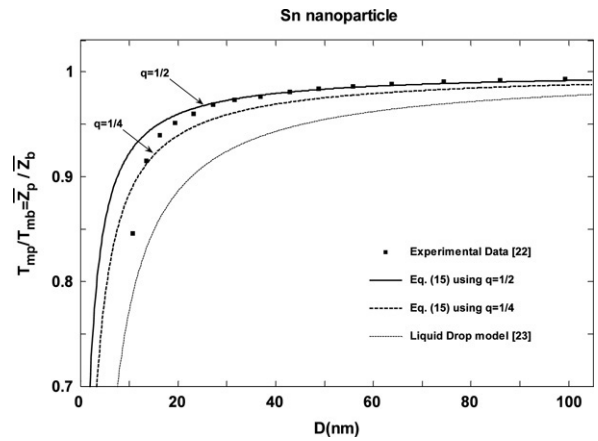

Fig. 5. Dependence of melting temperature on the size of Sn nanoparticles. Lattice type of Sn is BCT ($P_L = 0.53$, $P_S = 0.61$), $D_0 = 0.857$ nm (atomic radius from [23]) and $T_{mb} = 505.1$ K [23]. Experimental data from [22].

particles adequately large and close to the bulk, in our model is smaller than [23,24].

4. Conclusion

A model has been developed to calculate the average coordination number of nanoparticles by considering the coordination number in the surface and the lattice. We obtained a general equation for the average coordination number, which directly relates the average coordination number to the cohesive energy and the melting temperature of nanoparticles as a function of size. This relation shows that the ratio of the nanoparticles melting temperature to the bulk is equal to the ratio of nanoparticles average coordination number to the bulk average coordination number, as well as the ratio of particle cohesive energy to that of the bulk. This work has been compared with the Liquid Drop model and shows good consistency with reported experimental and MD simulations data.

References

[1] A.I. Frenkel, C.W. Hills, R.G. Nuzzo, J. Phys. Chem. B 105 (2001) 12689.
[2] G. Schmid, M. Baumle, M. Geerkens, I. Helm, C. Osemann, T. Sawitowski, Chem. Soc. Rev. 28 (1999) 179.
[3] P. Moriarty, Rep. Prog. Phys. 64 (2001) 297.
[4] J.M. Montejano-Carrizales, F. Aguilera-Granja, J.L. Moran-Lopez, Nanostruct. Mater. 8 (1997) 269.
[5] F. Baletto, R. Ferrando, Rev. Mod. Phys. 77 (2005) 371.
[6] S.H. Huh, H.K. Kim, J.W. Park, G.H. Lee, Phys. Rev. B 62 (2000) 2937.
[7] A.M. Stoneham, J. Phys.: Condens. Matter 11 (1999) 8351.
[8] C.T. Campbell, S.C. Parker, D.E. Starr, Science 298 (2002) 811.
[9] M.G. Mason, Phys. Rev. B 27 (1983) 748.
[10] F. Aguilera-Granja, S. Bouarab, A. Vega, J.A. Alonsoa, J.M. Montejano-Carrizalesb, Solid State Commun. 104 (1997) 635.
[11] S.C. Vanithakumari, K.K. Nanda, J. Phys. Chem. B 110 (2006) 1033.
[12] C.Q. Sun, B.K. Tay, X.T. Zeng, S. Li, T.P. Chen, J. Zhou, H.L. Bai, E.Y. Jiang, J. Phys.: Condens. Matter 14 (2002) 7781.
[13] C.Q. Sun, Y. Wang, B.K. Tay, S. Li, H. Huang, Y.B. Zhang, J. Phys. Chem. B 106 (2002) 10701.
[14] F. Lequien, J. Creuze, F. Berthier, B. Legrand, J. Chem. Phys. 125 (2006) 094707.
[15] D. Xie, M.P. Wang, W.H. Qi, J. Phys.: Condens. Matter 16 (2004) L401–L405.

[16] M. Schmidt, R. Kusche, B. von Issendorff, H. Haberland, Nature 393 (1998) 238.
[17] J.G. Dash, Rev. Mod. Phys. 71 (1999) 1737.
[18] Ph. Buffat, J.P. Borel, Phys. Rev. A 13 (1976) 2287.
[19] Ph. Buffat, Thin Solid Films 32 (1976) 283.
[20] E.A. Olson, M.Yu. Efremov, M. Zhang, Z. Zhang, L.H. Allen, J. Appl. Phys. 97 (2005) 034304.
[21] M. Wautelet, J. Phys. D 24 (1991) 343.
[22] S.L. Lai, J.Y. Guo, V. Petrova, G. Ramanath, L.H. Allen, Phys. Rev. Lett. 77 (1996) 99.
[23] K.K. Nanda, S.N. Sahu, S.N. Behera, Phys. Rev. A 66 (2002) 013208.
[24] K.K. Nanda, Chem. Phys. Lett. 419 (2006) 195.
[25] W.H. Qi, M.P. Wang, Mater. Chem. Phys. 88 (2004) 280.
[26] Q. Jiang, H.X. Shi, M. Zhao, J. Chem. Phys. 111 (1999) 2176.
[27] M. Attarian Shandiz, A. Safaei, S. Sanjabi, Z.H. Barber, J. Phys. Chem. Solids 68 (2007) 1396.
[28] A. Safaei, M. Attarian Shandiz, S. Sanjabi, Z.H. Barber, J. Phys.: Condens. Matter 19 (2007) 216216.
[29] M. Wautelet, Phys. Lett. A 246 (1998) 341.
[30] T.P. Martin, Phys. Rep. 273 (1996) 199.
[31] H.K. Kim, S.H. Huh, J.W. Park, J.W. Jeong, G.H. Lee, Chem. Phys. Lett. 354 (2002) 165.
[32] F. Cellestini, R.J.-M. Pellenq, P. Bordarier, B. Rousseau, Z. Phys. D 37 (1996) 49.